TravelNova — Brazil Alternative Payments

Technical Analysis & Action Plan

PIX & Boleto Approval Degradation — 34-Day Review (April 1 – May 4, 2024)

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Date: May 6, 2024 | Confidential — for executive review

Executive Summary

Brazil approval rates fell from **89.1% to 83.4% blended** over five weeks. Our analysis of 38,214 PIX attempts and 14,672 Boleto attempts shows this is **not a platform-wide failure and not a Yuno routing failure**: it is the collision of Ebanx's April 15 transaction-ID uniqueness enforcement with latent configuration defects in TravelNova's integration, plus a Boleto registration/payload misconfiguration that surfaced the same week.

Configuration errors account for **55.8% of PIX declines and 80.8% of Boleto failures**. Fixing them should return PIX to ~93.9% and Boleto to ~91%, recovering roughly **$1.3–1.9M per month** in gross booking value. Three fixes — merchant configuration repair, transaction-ID uniqueness, and Boleto registration completion — are already scoped with owners and timelines in Section 2.

1. Root Cause Analysis

1.1 What the data shows

Three facts frame the problem. First, the decline is **concentrated in PIX and Boleto**; card approval has been flat at ~78% for five weeks, which is normal for Brazil's high-fraud-screening card environment. Second, the inflection point is **Week 3 (Apr 15–21)** — exactly when Ebanx began enforcing transaction-ID uniqueness (Apr 15). Third, the degradation is **provider-specific**: Ebanx PIX dropped 8.0 pp (94.1% → 86.1%) while PagSeguro PIX dropped only 3.2 pp (92.9% → 89.7%) on the same traffic patterns. If Yuno's routing were broken, or TravelNova's platform were failing, both PIX providers, cards, and all Boleto issuers would degrade roughly in parallel. They did not.

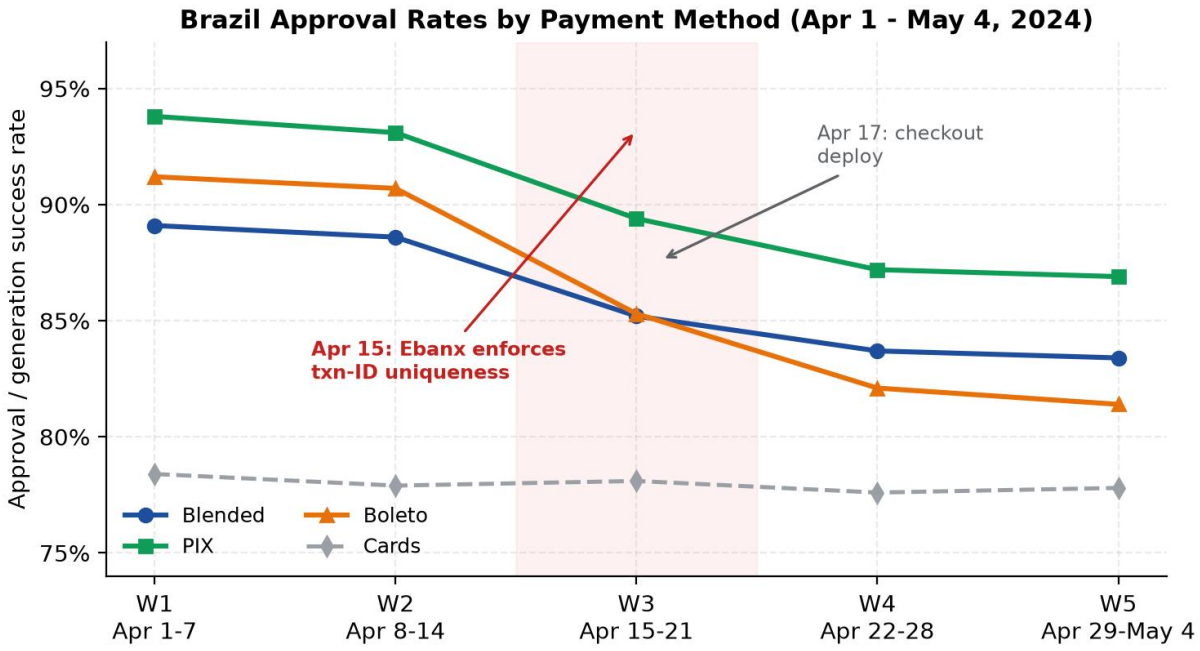


Figure 1 — Approval by payment method. The break occurs in Week 3, coinciding with Ebanx's Apr 15 enforcement.

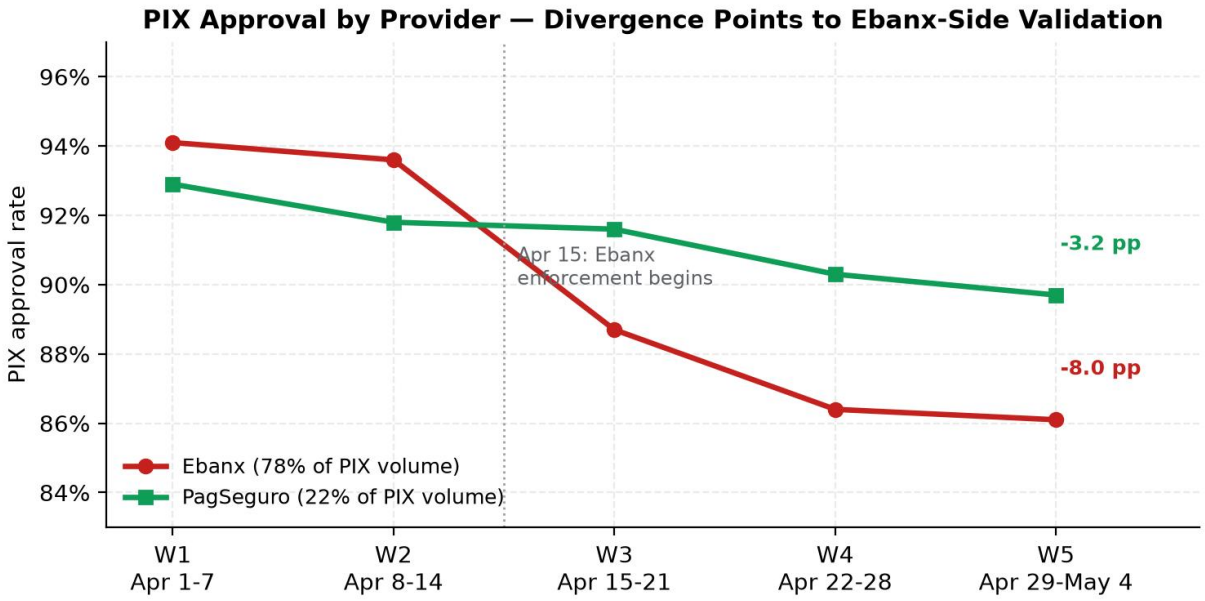


Figure 2 — PIX approval by provider. Ebanx (-8.0 pp) vs PagSeguro (-3.2 pp) divergence points to Ebanx-side validation exposure.

1.2 PIX: a validation change exposed latent integration defects

PIX approval measured over Apr 15 – May 4 is 86.15% (5,293 declines / 38,214 attempts), down from a 93.8% Week-1 baseline. The decline breakdown is decisive:

Code	Description	Count	Type	% of declines
PIX-03	QR generation failed — invalid merchant configuration	1,847	**Config error**	34.9%
PIX-33	Duplicate transaction ID detected (72h window)	1,104	**Config error**	20.9%
PIX-12	Timeout — customer did not complete in window	923	Soft decline	17.4%
PIX-08	Invalid PIX key format (customer-entered)	612	Hard decline	11.6%
PIX-21	Issuer temporarily unavailable	478	Soft decline	9.0%
PIX-17	Customer account restriction / daily limit	329	Soft decline	6.2%

Configuration errors (PIX-03 + PIX-33) total **2,951 declines — 55.8% of all PIX declines**. The arithmetic closes almost perfectly: relative to the 93.8% baseline, TravelNova accumulated 2,924 excess PIX declines over Apr 15 – May 4. The 2,951 config-error declines explain **99.1% of the entire excess**. In other words, if PIX-03 and PIX-33 are eliminated, PIX approval returns to ~93.9% — baseline performance.

The two config errors have distinct mechanisms:

- **PIX-03 — merchant configuration mismatch.** Ebanx's Apr 23 partner notice explicitly flagged this pattern: merchant configuration fields (e.g., merchant_account_id) not matching registered values. Ebanx's mid-April validation

tightening appears to have turned a previously tolerated mismatch into a hard failure. This is a registration/credential issue, not customer behavior.

- **PIX-33 — duplicate transaction IDs.** Ebanx's Apr 15 enforcement (Central Bank compliance) rejects any transaction ID duplicated within 72 hours. Two plausible sources must both be audited: (a) TravelNova's checkout reusing an ID across customer retries — potentially amplified by the Apr 17 React state refactor inadvertently double-invoking the payment call; and (b) retry logic — including the Yuno-managed Ebanx→PagSeguro fallback enabled Mar 28 — resubmitting the same ID. We own checking (b) and will publish results; daily PIX-33 counts before vs after Apr 17 will quantify (a).

1.3 Boleto: registration and payload defects across all issuers

All four issuer banks degraded almost identically (Banco do Brasil -10.2 pp, Itaú -10.2 pp, Bradesco -9.7 pp, Santander -9.5 pp from Week 1 to Week 5). Uniform degradation across issuers rules out an issuer-side problem; the defect sits in what TravelNova/Ebanx sends to every issuer:

Code	Description	Count	Type	% of failures
BOL-14	Merchant registration incomplete for issuer	2,917	Config error	43.0%
BOL-06	Invalid expiration date format (must be YYYY-MM-DD)	1,523	Config error	22.5%
BOL-19	Duplicate document number	1,038	Config error	15.3%
BOL-22	Customer CPF validation failed	891	Hard decline	13.1%
BOL-11	Amount exceeds issuer limit (>R$5,000)	412	Hard decline	6.1%

Configuration errors (BOL-14 + BOL-06 + BOL-19) are **80.8% of itemized Boleto failures**. BOL-14 (registration incomplete per issuer) mirrors the PIX-03 pattern and strongly suggests the same underlying configuration drift — most plausibly exposed by Ebanx tightening validations in mid-April. BOL-06 is a date-format defect in the payload, and BOL-19 indicates document-number generation is reusing values, echoing the PIX-33 uniqueness problem. **Data caveat:** the itemized table (6,781 failures / 14,672 attempts = 46.2%) is inconsistent with the weekly success tables, which imply an 8.8%–18.6% failure rate over the same period. The itemized table likely includes retries and/or the attempt base is undercounted. We therefore quantify impact conservatively from the weekly rate delta, not the itemized counts.

1.4 The April 17 deploy: correlation, not (yet) proven cause

The checkout deploy is the obvious suspect, but the evidence does not support it as the primary driver:

- **Timing:** PIX and Boleto both began deteriorating in Week 3, and Ebanx's enforcement started Apr 15 — two days before the deploy. A frontend-only change cannot explain Boleto registration errors (BOL-14) at all.
- **Segmentation:** PagSeguro PIX held at ~90% and cards held at ~78% throughout. A platform-wide bug from the deploy would not spare PagSeguro traffic routed through the same checkout.
- **Where the deploy may matter:** a React state-management refactor can silently introduce duplicate API invocations (e.g., re-render double-fire or a retry-on-unmount effect). If daily PIX-33 volumes step up after Apr 17, the deploy is an amplifier of the duplicate-ID problem — even though the root trigger was Ebanx's policy. We have requested daily PIX-33 counts and Ricardo's frontend call logs to settle this within 24 hours.

Honest bottom line: Ebanx's policy change is the trigger; TravelNova-side configuration drift (and possibly duplicate frontend calls) is the fuel. Neither Yuno routing nor a general provider outage is consistent with the data.

1.5 Revenue impact

Volume math: $47M/month total × 58% Brazil = $27.26M/month. PIX = 62% ($16.90M), Boleto = 23% ($6.27M), Cards = 15% ($4.09M).

Driver	Excess failures / month	Gross booking value at risk	Basis
PIX-03 (merchant config)	2,770	**$817K**	1,847 × 30/20 × $295 avg ticket
PIX-33 (duplicate txn IDs)	1,656	**$488K**	1,104 × 30/20 × $295
Boleto config errors (BOL-14/06/19)	2,157	**$614K**	(91.2% − 81.4%) × 22,008 attempts × $285
Total (fixable config defects)	—	**$1.91M / month**	≈ 4.1% of total GMV

Not every failed attempt is a lost booking — some customers retry or switch method. Assuming ~25% self-recovery, net monthly revenue at risk is approximately **$1.4M**. Soft PIX declines (PIX-12/21/17, ~$765K/month in attempted value) are a secondary recovery opportunity through retry logic and UX (Section 2, action 5). These figures are consistent throughout Sections 2–4.

2. Technical Action Plan

Actions are prioritized by revenue impact. P0 items start today; all P0s are scoped to complete before Wednesday's executive review or within 48 hours of approval.

#	Action	Owner	Expected impact	Timeline	Priority
1	Fix Ebanx merchant configuration: reconcile merchant_account_id and all registered config fields against Ebanx's records (PIX-03); apply corrected credentials in Yuno vault and re-run certification suite	Yuno + Ebanx (TravelNova provides final account confirmation)	**PIX +4.8 pp (86.2% → ~91%); ~$817K/mo**	Today — Tue EOD	**P0**
2	Eliminate duplicate transaction IDs (PIX-33): (a) audit Yuno fallback retry path to regenerate provider txn ID on every attempt; (b) TravelNova engineering audits ID generation across customer retries and adds idempotency keys to the checkout API call; (c) diff daily PIX-33 counts pre/post Apr 17 deploy to quantify frontend double-fire contribution	Yuno (a) + TravelNova engineering (b, c)	**PIX +2.9 pp; ~$488K/mo**	a: today; b/c: this week	**P0**
3	Complete Boleto merchant registration per issuer (BOL-14) with Ebanx; fix expiration-date payload to YYYY-MM-DD (BOL-06); regenerate document-number sequence (BOL-19)	TravelNova + Ebanx, Yuno facilitates	**Boleto toward ~91% baseline; ~$614K/mo**	This week	**P0**
4	Tune PIX routing: temporarily shift Ebanx/PagSeguro split from 78/22 toward 60/40 while fixes land; exclude config-error codes from retry eligibility (retrying a config error always fails); auto-fallback PIX-21 issuer-unavailable to PagSeguro	Yuno	**+1.0 pp blended PIX ≈ $169K/mo interim lift**	Immediate (config change)	**P1**
5	Soft-decline recovery: extend PIX-12 QR expiry window and add expiry reminder; enable smart retry for PIX-17 after limit reset; A/B test retry UX	Yuno + TravelNova (UX)	Partial recovery of ~$765K/mo attempted value; est. +1–2 pp PIX	Next sprint	**P1**

#	Action	Owner	Expected impact	Timeline	Priority
6	Prevent recurrence: payment-telemetry dashboards with error-code-mix alerting (any code >2% of declines pages both teams), weekly provider scorecard, quarterly config-drift audit of all merchant credentials vs provider registrations	Yuno (tooling) + TravelNova (on-call integration)	**Detects this class of issue in hours, not 19 days**	Next sprint	P2
7	Joint RCA review: publish final Ebanx policy-impact report, including deploy-contribution findings from action 2c; agree prevention playbook	All three parties	Closes the loop for exec review; documents accountability	Friday May 10	P2

2.1 Expected recovery

With actions 1–3 complete, PIX returns to ~93.9% (config errors fully removed from the Apr 15 – May 4 baseline of 86.15%), Boleto returns to ~91%, and blended Brazil approval recovers from 83.4% to ~89% — approximately $1.3–1.9M/month in gross booking value restored, directly supporting H2 GMV targets. Validation: 48-hour post-fix error-code audit confirming PIX-03/PIX-33 and BOL-14/06/19 at <0.5% of declines, plus daily approval tracking against the Week-1 baseline.

3. Merchant-Facing Communication

Email to Daniela Rocha, Monday May 6, 2024 (292 words):

Subject: Brazil payments — root cause identified, 3-step fix underway

To: Daniela Rocha | Cc: Ricardo Mendes

Hi Daniela,

We've completed our analysis of Brazil's PIX and Boleto decline, and I want to give you clear answers before Wednesday.

What happened: On April 15, Ebanx began strictly enforcing transaction-ID uniqueness (a Central Bank of Brazil compliance requirement) and tightened configuration validation. This turned small mismatches in the integration — some on our orchestration side, some in configuration registered with Ebanx — into hard payment failures. The result: configuration issues now cause 56% of all PIX declines and 81% of Boleto failures.

What it's costing you: Roughly $1.4M–$1.9M per month in gross bookings (about 4% of total GMV). PIX is at 86% approval versus your 94% first-quarter baseline; Boleto at 81% versus 91%.

What we're doing about it: (1) Today: correcting the Ebanx merchant configuration causing most PIX failures. (2) This week: eliminating duplicate transaction IDs and completing Boleto registration with each issuing bank. (3) Immediately: shifting more PIX volume to your backup provider (which has held a 90% approval rate throughout) while fixes land. Our data shows this returns PIX to ~94% and Boleto to ~91% within days of deployment.

To be direct about causality: this was not a Yuno routing failure — your card payments and backup PIX provider were stable throughout, which is exactly what the routing layer protected you from. It's also why fixing this direct-to-provider would be harder: you'd be debugging Ebanx's validation changes across three payment methods alone.

What we need from your team: Ricardo's engineers for a 1-hour session by Tuesday EOD to verify transaction-ID generation and add a duplicate-call guard to the new checkout. We'll bring everything else.

I'll send confirmation as each fix lands. Happy to join Wednesday's review if useful.

Best,

[Name] — Technical Account Manager, Yuno

4. Proactive Optimization Recommendations

Beyond the immediate fixes, three opportunities align with TravelNova's Series B growth targets and would materially strengthen Brazil's payment economics:

4.1 Rebalance PIX routing and renegotiate on data (routing + cost)

PagSeguro has outperformed Ebanx by 3.6 pp in Week 5 (89.7% vs 86.1%), and even pre-incident the gap was ~1–2 pp with Ebanx handling 78% of volume. Once the config fixes land, we recommend a **4-week A/B routing test at a 50/50 split**. Every +1 pp of PIX approval is worth ~$169K/month at current volume. The same dataset doubles as negotiating leverage in your next Ebanx commercial review — performance-based volume commitments in exchange for improved pricing.

4.2 Add a second Boleto provider and solve the high-value ceiling (resilience + conversion)

Boleto runs 100% through Ebanx — a single point of failure this incident just demonstrated. Adding a backup Boleto provider (e.g., via PagSeguro or a direct bank integration) costs one connector through Yuno and removes that dependency. Separately, BOL-11 failures (~618/month, ~$176K in attempted value) are high-value bookings above the R$5,000 issuer limit: we recommend automatically splitting these into two vouchers or routing them to card installments — recovering bookings that today fail outright and improving the experience exactly where average order value is highest.

4.3 Lift card approval from 78% toward 83–85% (largest untapped headroom)

Cards are stable but flat at 78%, below the 82–85% that well-tuned Brazil card flows achieve. Three levers: (a) **3DS2 optimization** — dynamic triggers instead of blanket challenges; (b) **smart retry on soft declines** (insufficient funds, limits) with 24-hour re-attempt; (c) **BIN-level routing** between Adyen and Stone by issuer performance. A conservative +3 pp is worth ~$123K/month on $4.09M card volume — and unlike the current incident, this is pure upside, not recovery.

Recommended sequencing: Boleto redundancy in the next sprint (resilience while the incident is fresh), the PIX A/B test immediately after config fixes validate (week of May 20), and the card optimization program as a standing quarterly initiative. Together these are worth an estimated **$450K–$700K/month** in incremental approvals — the difference between firefighting and compounding.

Sources: TravelNova Brazil transaction data Apr 1 – May 4, 2024 (weekly approval tables; PIX decline breakdown, 38,214 attempts; Boleto decline breakdown, 14,672 attempts); Yuno Support Tickets #8821, #9214; Ebanx partner communications Apr 10 / Apr 23; TravelNova deploy log Apr 17. Revenue figures are monthly gross booking value estimates; Boleto impact derived from weekly rate deltas due to the data inconsistency documented in Section 1.3.